COMMUNITY & LIVING • SMART CITY INNOVATION

N-SHIELD

AI-Powered Urban Safety Intelligence

A Citizen-Centric Smart Safety Platform for Nagpur, Maharashtra



PROJECT LEAD & DEVELOPER
Khushal Deogade

TARGET CITY
Nagpur, Maharashtra, India

SUBMISSION TARGET
Academic & Hackathon / Fund My Crazy

DOCUMENT VERSION
v1.0 Complete Specification

2 Abstract / Project Summary

N-SHIELD (AI-Powered Urban Safety Intelligence) is a comprehensive, citizen-focused smart city safety platform specifically architected for Nagpur, Maharashtra. The system synthesizes spatial map visuals, verified public-safety updates, emergency services directory data, and artificial intelligence into a single unified interactive dashboard. By consolidating fragmented safety data into intuitive visual indicators and plain-language insights, N-SHIELD empowers citizens to evaluate neighborhood risk profiles before traveling, choosing commute routes, visiting unfamiliar areas, or responding to emergency situations.

> **Core Value Proposition:** Transforming fragmented, scattered public safety signals into proactive, actionable spatial intelligence for everyday citizens without creating panic or spreading unverified rumors.

In rapidly growing tier-1 and tier-2 smart cities across India such as Nagpur, urban expansion has outpaced the delivery of localized, accessible safety context. While navigation platforms excel at providing shortest-path routing based on traffic congestion, they remain completely blind to safety variables such as lighting conditions, localized crime trends, nearby emergency support density, and active police patrol zones. N-SHIELD bridges this critical gap.

Key Core Pillars of N-SHIELD

- **Spatial Risk Map:** Color-coded visual overlay (Green, Orange, Red) representing relative safety levels across major commercial, residential, and transit zones in Nagpur.
- **Emergency Support Directory:** One-tap access to nearby Police Stations, Hospitals, 24/7 Medical Stores, and Women's Helpline contacts filtered by current GPS location.
- **Verified Safety News Feed:** Real-time aggregation of local news, municipal alerts, and police advisories verified to prevent misinformation.
- **Responsible AI Analytics:** Powered by Gemini API to generate plain-language safety summaries and practical situational advice for commuters.
- **Scalable Architecture:** Built on modern full-stack web standards (Next.js, TypeScript, Google Maps API, MongoDB/PostgreSQL) designed for seamless expansion across Maharashtra and India.

3 Problem Statement

Urban safety in Indian cities presents a complex challenge characterized by fragmented information access, lack of spatial safety context, and emergency response friction. Citizens navigating Nagpur daily face five distinct structural pain points:

Problem Area	Current Reality in Nagpur	Impact on Citizens
1. Fragmented Safety Data	Crime updates, emergency helplines, police station locations, and hospital contacts are scattered across disparate websites, PDF documents, news portals, and social media feeds.	Citizens lose vital minutes during emergencies searching for phone numbers or verified locations.
2. Blind Spot Navigation	Standard navigation maps optimize solely for travel distance and traffic time. They provide zero visibility into low-lit stretches, isolated streets, or high-incident junctions.	Commuters (especially women and night-shift workers) unknowingly take risky or poorly patrolled routes.
3. Lack of Contextual Awareness	Citizens traveling to unfamiliar neighborhoods (e.g., late-night arrivals at Nagpur Railway Station or Mihan IT Hub) have no quick way to assess area safety.	Increased vulnerability to petty crime, theft, harassment, or delayed assistance.
4. Rumors & Misinformation	Unverified social media posts often cause unnecessary panic regarding local safety incidents or road closures.	Public anxiety increases while genuine safety warnings get lost in digital noise.
5. High Bar to Data Utilization	Raw crime statistics published by official crime bureaus are technical, tabular, and difficult for non-technical citizens to interpret in daily life.	Valuable safety statistics remain unused by the very citizens who need actionable advice.

4 Proposed Solution

N-SHIELD solves these systemic challenges by introducing a unified, intelligent urban safety web platform engineered with citizen usability at its core. Rather than presenting intimidating statistical tables, N-SHIELD transforms safety data into intuitive visual layers and actionable AI summaries.

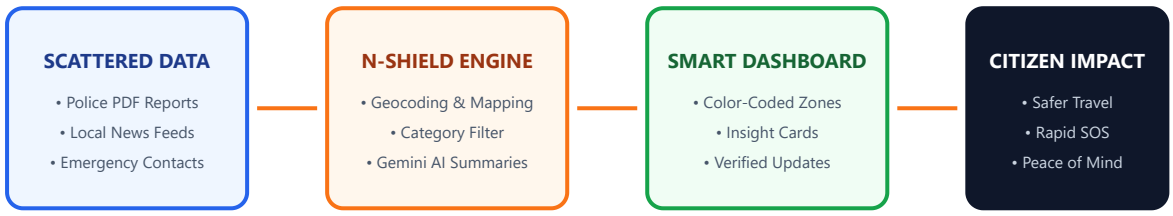


Figure 4.1: N-SHIELD Core Transformation Pipeline

Core Solution Components

- **Interactive Nagpur Safety Map:** Embedded Google Maps interface with dynamic risk overlay polygons and location marker clustering for Nagpur landmarks (Sitabuldi, Dharampeth, Sadar, Wardhaman Nagar, Hinna Road, Mahal, Mihan, Nagpur Station).
- **Color-Coded Risk Zones:**
  - GREEN ZONE (LOW RISK) Well-lit, highly patrolled commercial/residential areas with active public presence.
  - ORANGE ZONE (MEDIUM RISK) Areas requiring standard caution, high-density traffic junctions, or moderate past incidents.
  - RED ZONE (HIGH RISK / CAUTION) Isolated stretches, unlit roads, or areas with frequent recent reported incidents.
- **Multi-Category Filters:** User-controlled filters to isolate specific safety parameters (Theft, Robbery, Harassment, Traffic Accidents, Unlit Stretches).
- **Nearby Essential Services Finder:** Instant spatial search for nearby Police Stations, Hospitals, and 24/7 Pharmacies with direct click-to-call links and navigation routes.
- **AI-Powered Safety Insight Card:** Gemini AI summarizes public safety signals into concise, non-alarmist bullet points and safety tips.

IMPORTANT SAFETY GUIDANCE & ETHICAL DISCLAIMER

N-SHIELD is designed exclusively to provide general safety awareness and public information guidance. It does *NOT* claim to predict individual crime events, guarantee absolute safety, or replace official law enforcement and emergency services. In cases of immediate danger or threat to life, citizens must immediately call national emergency services (112 / 100). All risk indicators represent historical and aggregated spatial public patterns, not individual profiling.

5 Objectives

Primary & Secondary Goals

1. **Enhance Citizen Safety Awareness:** Equip residents and visitors of Nagpur with real-time, neighborhood-specific safety data before they travel.
2. **Frictionless Emergency Access:** Reduce emergency response connection time by providing one-click dialing to nearest police stations and medical centers.
3. **Centralize Fragmented Data:** Combine maps, news alerts, municipal data, and hospital directories into one seamless, responsive web application.
4. **Promote Responsible AI Usage:** Demonstrate how AI models can summarize complex public incident reports without hallucinating, biased profiling, or sensationalism.
5. **Enable Data-Driven Decisions:** Assist commuters, night workers, and travelers in choosing safer travel times and better lit transit corridors.
6. **Build a Scalable Smart City Template:** Establish a modular technology architecture that can be cloned and deployed across other smart cities in India (e.g., Pune, Nashik, Mumbai, Thane).

6 Target Users & Stakeholder Analysis

N-SHIELD is architected to support a diverse spectrum of urban citizens, visitors, and municipal stakeholders in Nagpur. The matrix below outlines how specific user groups interact with and benefit from the platform:

User Segment	Primary Pain Points / Needs	How N-SHIELD Helps
Students *(RKNEC, VNIT, HISLOP, Nagpur Univ.)*	Late evening coaching classes, late library hours, traveling alone on two-wheelers, unfamiliar hostel surroundings.	Provides clear risk zone guidance, nearby police station finder, and instant emergency call buttons for peace of mind while commuting.
Women Commuters *(IT Professionals, Retail Staff)*	Night-shift travel from MIHAN IT Park, poorly lit bus stops, unpatrolled suburban routes, feeling unsafe at isolated auto stands.	Delivers real-time location insights, safety tips for specific areas, verified news alerts, and safe route guidance.
Working Professionals *(Late Shift Workers, Delivery Drivers)*	Navigating unfamiliar routes during late night hours, finding open 24/7 medical stores or urgent care facilities.	Quick location search with hospital & pharmacy locator, real-time risk status, and direct Google Maps navigation integration.
Senior Citizens	Difficulty navigating complex apps, finding verified emergency helpline numbers, assessing neighborhood safety for evening walks.	Simplified high-contrast user interface, large click-to-call emergency buttons, plain-language AI safety summaries.
Tourists & Visitors *(Nagpur Railway / Airport Arrivals)*	No prior knowledge of Nagpur geography, safe areas, or local police station boundaries.	Interactive safety map highlighting major hubs (Sitabuldi, Sadar) with immediate clarity on safe commercial zones vs caution areas.
Families	Concern for children returning home late, selecting safe residential neighborhoods for home buying or renting.	Aggregated neighborhood safety trends, verified community safety news, and downloadable area safety insights.
Police & Municipal Authorities *(Future Version Scope)*	Need for aggregated community feedback on unlit streetlights, dead spots, and citizen safety perception patterns.	Provides anonymous crowdsourced safety feedback metrics to optimize police patrolling routes and civic lighting infrastructure.

7 Key Features Deep-Dive

Detailed Feature Breakdown

1. Interactive Google Maps Safety Layer: Custom Google Maps JS integration displaying Nagpur color-coded safety overlays, heatmaps, and custom markers. Smooth zoom, pan, and tilt controls tailored for both desktop and mobile screens.

2. Intelligent Location Search & Geocoding: Integrated Google Places Autocomplete search bar enabling instant location lookup across all major Nagpur localities (e.g., Dharampeth, Sadar, Itwari, Wardha Road, Hingna).

3. Dynamic Risk-Zone Visualization: Real-time rendering of spatial boundaries categorized into Green (Low Risk), Orange (Medium Risk), and Red (Caution/High Risk) based on historical incident density and lighting conditions.

4. Safety Category Filters: Toggle switches allowing users to filter map layers by specific concern types: Theft & Robbery, Harassment, Traffic Hazards, Unlit Stretches, and Low Patrol Zones.

5. Location-Based Safety Insight Card: Dynamic sidebar/bottom-sheet panel updating automatically when a location is clicked. Displays overall risk score (1-100), primary risk factors, and recommended safety precautions.

6. Nearby Police Station Finder: Spatial database query identifying nearest police stations relative to the user's current GPS location or selected map pin, displaying official contact numbers, address, and distance.

7. Nearby Hospitals & 24/7 Pharmacy Finder: Live locator for emergency trauma centers, multi-specialty hospitals, and late-night medical stores with direct call buttons and driving directions.

8. One-Tap Emergency SOS Call Suite: Prominent floating emergency toolbar providing immediate click-to-call access to National Emergency (112), Police Hotline (100), Women's Helpline (1091), Ambulance (102), and Fire (101).

9. Verified Local Safety News & Alert Feed: Curated feed of verified municipal advisories, traffic diversions, weather warnings, and local safety updates sourced from verified news feeds and police handles.

10. Gemini AI Safety Summarizer: Advanced LLM prompt pipeline that converts raw news articles and multi-point incident data into concise 3-bullet plain-language summaries without hype or sensationalism.

11. Location-Specific AI Safety Tips: Context-aware advisory engine delivering tailored safety recommendations based on time of day (e.g., "Avoid unlit bypass road after 10 PM; stick to main Wardha Road corridor").

12. Turn-by-Turn Navigation Launchers: One-click deep-linking into Google Maps native application for instant navigation to selected safe locations or emergency centers.

13. Responsive Cross-Platform PWA Layout: Fluid, modern mobile-first user interface built with Next.js and Tailwind CSS, fully optimized for Android devices, iPhones, tablets, and desktop browsers.

14. Safer Route Comparison (Future Scope Ready): Architectural hooks for evaluating alternative origin-to-destination routes by overlaying risk zone polygons onto Google Directions API paths.

8 Complete User Workflow

The end-to-end user workflow is engineered to deliver critical safety context within 3 clicks of opening the application. Below is the visual workflow architecture:



Figure 8.1: Complete End-to-End User Interaction Flow

9 Technology Stack

N-SHIELD is built using modern, production-grade open-source web frameworks, cloud mapping services, scalable database models, and generative AI APIs. Below is the detailed technical breakdown:

Component	Technology Chosen	Version / Spec	Engineering Rationale & Benefits
Frontend Framework	Next.js (React)	v14+ (App Router)	Server-Side Rendering (SSR) for fast initial load, optimized image loading, built-in API routing, and excellent SEO capabilities.
Programming Language	TypeScript	v5.0+	Strict static typing eliminates runtime reference errors, improves IDE autocompletion, and enforces robust data contracts.
UI Styling & Theme	Tailwind CSS + Vanilla CSS	v3.4+	Utility-first CSS framework enabling rapid, highly customizable responsive UI components with clean dark/light modern themes.
Mapping Engine	Google Maps JS API	v3.55+	Industry-standard mapping solution offering high-precision spatial rendering, custom overlay polygons, geocoding, and smooth interaction.
Directions Engine	Google Directions API	v1 REST API	Calculates precise distance, duration, and turn-by-turn navigation paths for future safer-route comparisons.
Backend API Layer	Next.js API Routes / Node.js	ES2023 / Node 20	Lightweight serverless API functions co-located with frontend code, ensuring low latency and frictionless Vercel deployment.
Alternative Backend	FastAPI (Python) / Spring Boot	Python 3.11+	Modular Python API option for heavy data analysis, spatial geospatial calculations (GeoPandas), and ML model hosting.
Primary Database	MongoDB / PostgreSQL	Mongo v7 / Postgres 16	MongoDB for flexible document storage of news & incidents; PostgreSQL + PostGIS extension for high-performance spatial queries.
AI Summarization Engine	Google Gemini API	Gemini 1.5 Flash	Ultra-fast, low-cost generative AI model ideal for real-time text summarization, safety advice synthesis, and structured JSON parsing.
News & Public Feeds	RSS Feeds / Custom Scrapers	JSON / XML APIs	Fetches verified public safety advisories, municipal announcements, and local RSS news channels.

Component	Technology Chosen	Version / Spec	Engineering Rationale & Benefits
Deployment & Hosting	Vercel Cloud Platform	Edge Network	Global CDN edge network providing automated CI/CD builds from GitHub, SSL certificates, zero downtime, and instant scaling.
Version Control	Git & GitHub	Git 2.40+	Distributed version control, branch management, issue tracking, and automated deployment triggers.

10 System Architecture

N-SHIELD adopts a multi-tier, decoupled architecture separating client presentation, spatial mapping, backend services, database persistence, and external AI processing.

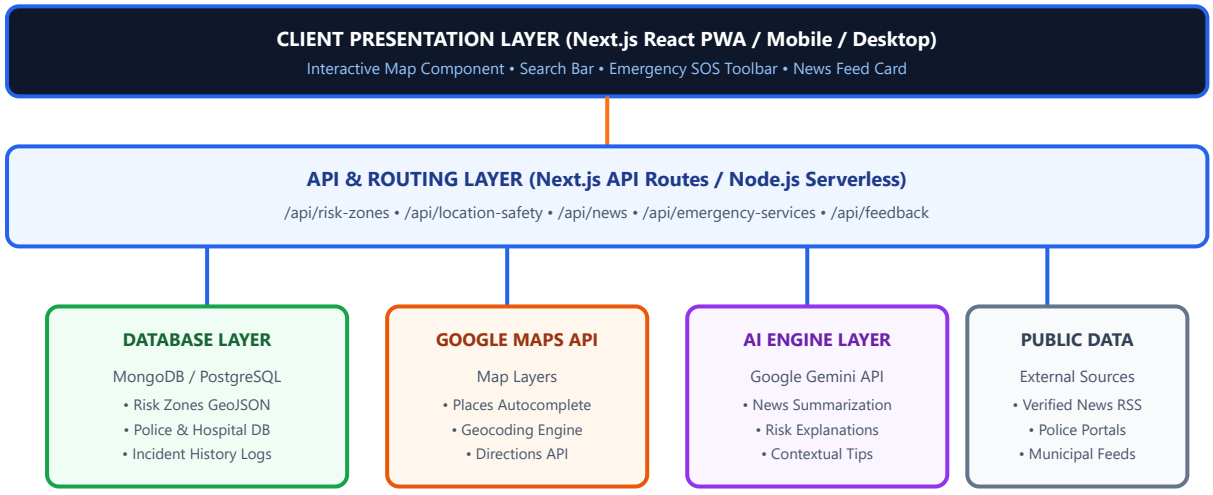


Figure 10.1: Multi-Tier N-SHIELD System Architecture

11 Data Flow & Privacy Framework

Step-by-Step Data Execution Sequence

1. **User Location Selection:** User grants browser GPS permission or types a specific location into the search bar (e.g., "Sitabuldi, Nagpur").
2. **Client Request Generation:** Next.js frontend dispatches asynchronous HTTPS requests to `/api/location-safety?lat=21.1458&lng=79.0882`.
3. **Backend Spatial Query:** API layer queries PostgreSQL/MongoDB spatial indexes to extract overlapping risk polygons, recent incident records within a 2 km radius, and nearest emergency service nodes.
4. **AI Processing Pipeline:** Extracted incident text and news headlines are passed to Google Gemini API with a structured prompt requesting a 3-bullet plain-language summary and safety advice.
5. **Client Rendering:** API returns compiled JSON payload; frontend updates map overlays, risk score gauge, emergency service lists, and AI summary card in real-time.

STRICT CITIZEN PRIVACY & DATA PROTECTION RULES

- **Zero Personally Identifiable Information (PII) Storage:** N-SHIELD does not require user registration or log individual tracking histories.
- **Transient Location Processing:** GPS coordinates received from user devices are evaluated in-memory solely to query nearby services and are never stored in databases or server logs.
- **No Commercial Data Tracking:** Zero third-party ad pixels, user fingerprinting, or data sales monetization models.
- **HTTPS Encrypted Payload Transport:** All client-to-server and server-to-API communication is strictly encrypted using TLS 1.3 standards.

12 AI Component & Responsible AI Framework

N-SHIELD leverages Google's Gemini 1.5 Flash API to translate complex, multi-source spatial safety data into clear, accessible guidance.

AI Functional Architecture

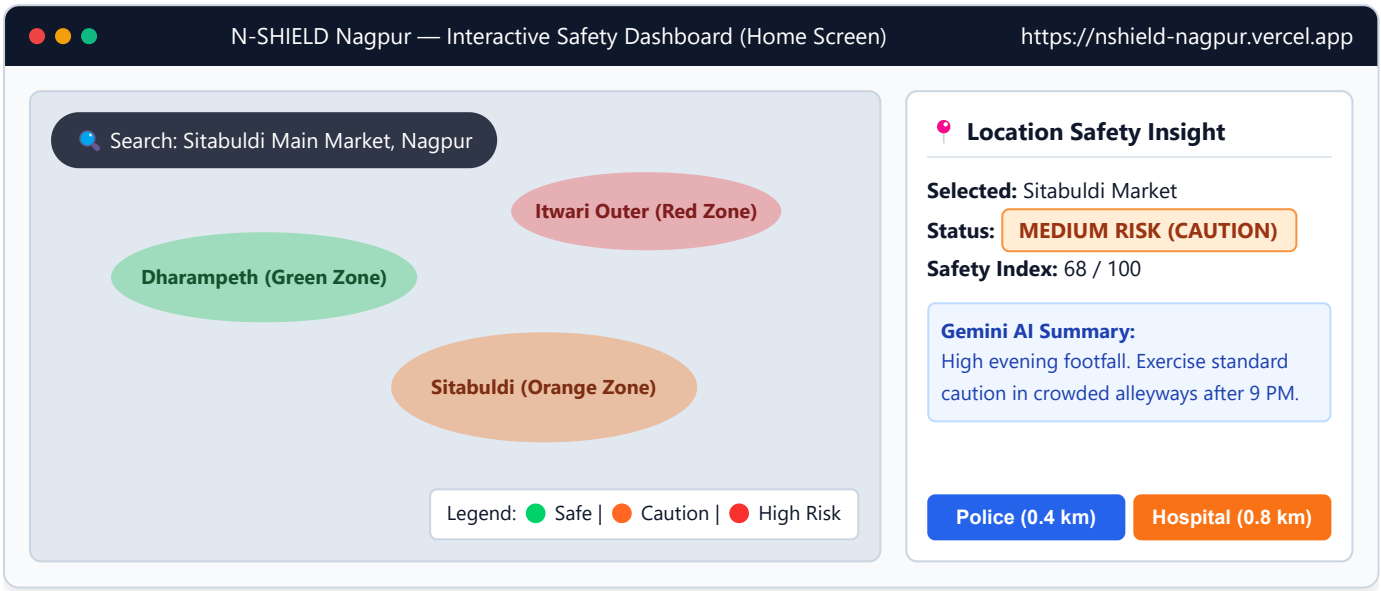
- **News & Advisory Summarization:** Distills lengthy news articles or municipal safety bulletins into 2-3 sentence summaries focusing strictly on practical safety implications.
- **Contextual Safety Tip Generation:** Dynamically formulates recommendations based on situational parameters (e.g., area risk level, current time of day, lighting availability).
- **Non-Technical Explanations:** Replaces intimidating crime statistics with plain-language context (e.g., instead of "Incident Rate Index 4.2", displaying "Moderate evening traffic; well-patrolled near main market").

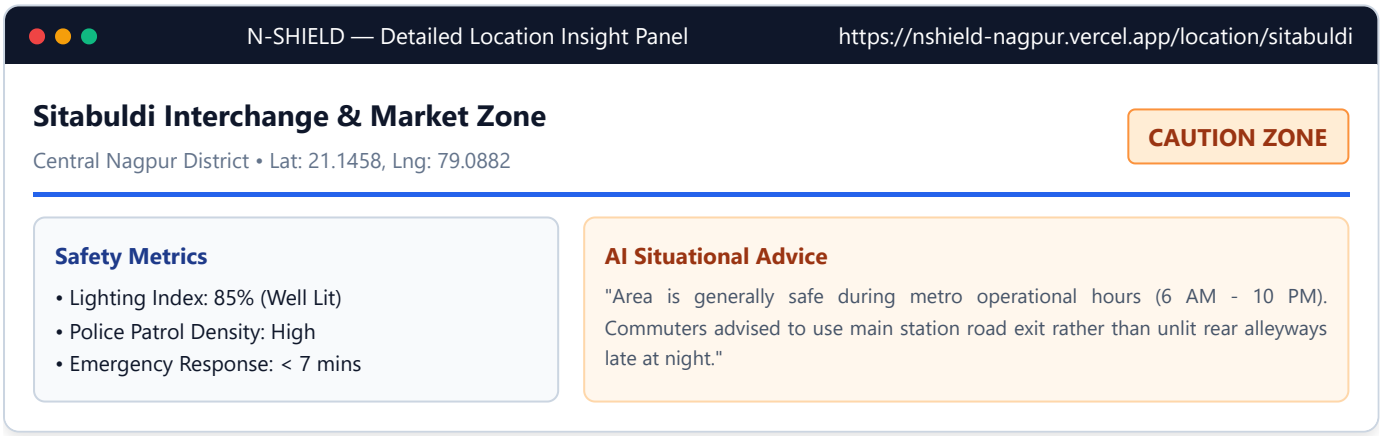
RESPONSIBLE AI GOVERNANCE RULES

Governance Rule	Enforcement & Technical Constraint
1. Zero Facial Recognition	Strictly prohibited. System processes zero biometrics, photos, or video streams.
2. No Individual Surveillance	No tracking of individual citizens, vehicles, or personal movement profiles.
3. No Personal Crime Prediction	System does not predict whether a specific individual will commit or be a victim of crime.
4. Anti-Bias Guardrails	Algorithms are strictly forbidden from utilizing demographic, religious, caste, gender, or income attributes in risk calculations.
5. Verified Source Requirement	AI models summarize only verified public news feeds, municipal portals, and police press releases.
6. Source Transparency	Every AI-generated insight clearly displays its verification status and underlying news source timestamp.

13 UI Pages & Screen Mockups (1 of 3)

Below are pixel-accurate layout blueprints representing key screens within the N-SHIELD application:





13 UI Pages & Screen Mockups (2 of 3)

N-SHIELD — Multi-Category Filter Interface

Filter Map Layers by Concern Category:

✓ ALL CATEGORIES | THEFT & ROBBERY | UNLIT STRETCHES | HARASSMENT HOTSPOTS | POLICE PATROL ZONES

[Active Map Layer: Displaying 14 verified incident points and 3 patrol routes matching selected criteria]

N-SHIELD — Verified Local News & Safety Advisories

Nagpur City Police Advisory — Today, 14:20 PM • Verified ✓

Special Security Patrols Deployed Near Sitabuldi Metro Station

Additional police personnel stationed to assist night commuters during festive season.

Municipal Corporation Alert — Yesterday, 19:45 PM • Verified ✓

Streetlight Maintenance Work Completed on Wardha Road Stretch

Illumination restored along 3.2 km stretch near Chhatrapati Square flyover.

N-SHIELD — Nearby Police & Hospital Emergency Finder

Sitabuldi Police Station

Distance: 0.4 km • Open 24 Hours

Call: 0712-2522222

GMC Hospital & Trauma

Distance: 1.2 km • ICU Available

Call: 0712-2744444

13 UI Pages & Screen Mockups (3 of 3)

N-SHIELD — One-Tap Emergency SOS Call Suite

🚨 DIRECT EMERGENCY CLICK-TO-CALL HOTLINES

📞 112 National | 👮 100 Police | 👩 1091 Women | 🚑 102 Ambulance

N-SHIELD — Safer Route Recommendation Engine (Future Scope Preview)

Origin: VNIT Campus ➔ Destination: Nagpur Railway Station

RECOMMENDED SAFE ROUTE

Via Wardha Road Main Arterial

12.4 km • 24 mins • 95% Illumination

✓ Continuous Police Patrols & CCTV

SHORTEST ROUTE (USE CAUTION)

Via Sub-Arterial Bypass

10.8 km • 21 mins • 40% Illumination

⚠ Unlit Stretches reported after 9 PM

N-SHIELD — Admin Moderation & Crowdsourced Safety Feedback Portal

Pending Feedback Reports: 12 Reports **News Verification Status:** 4 Feeds Active **System Status:** OPERATIONAL

Report ID	Location	Category	Details	Action
#FB-902	Hingna T-Point	Broken Streetlight	Entire flyover stretch dark	Verify & Update Map

14 Database Design & Schema Specifications

N-SHIELD utilizes a hybrid spatial document and relational data structure (MongoDB / PostgreSQL with PostGIS). Below are the core schemas defining the system entities:

1. Risk Zones Table / Collection (`risk_zones`)

Field Name	Data Type	Constraints	Description & Business Rules
`_id / id`	ObjectId / UUID	PRIMARY KEY	Unique identifier for the risk zone polygon.
`zone_name`	String	NOT NULL	Name of locality (e.g., "Sitabuldi Market Zone").
`city`	String	DEFAULT 'Nagpur'	City jurisdiction for scaling across India.
`risk_level`	Enum	'GREEN'|'ORANGE'|'RED'	Calculated safety rating tier.
`risk_score`	Integer	1 TO 100	Quantitative safety index score.
`boundary_geojson`	GeoJSON Polygon	SPATIAL INDEX	Coordinate polygon defining spatial area boundary.
`primary_factors`	Array[String]	NULLABLE	Key risk factors (e.g., ["unlit_roads", "high_crowd"]).
`updated_at`	Timestamp	NOT NULL	Timestamp of last automated calculation.

2. Locations Master Table (`locations`)

Field Name	Data Type	Constraints	Description
`location_id`	UUID	PRIMARY KEY	Unique location ID.
`name`	String	NOT NULL	Locality name (e.g., "Dharampeth").
`latitude`	Float / Decimal	-90.0 TO 90.0	Geographic latitude coordinate.
`longitude`	Float / Decimal	-180.0 TO 180.0	Geographic longitude coordinate.
`police_station_id`	UUID	FOREIGN KEY	Assigned police station jurisdiction.
`created_at`	Timestamp	DEFAULT NOW()	Record creation timestamp.

3. Police Stations Directory Table (`police_stations`)

Field Name	Data Type	Constraints	Description
`station_id`	UUID	PRIMARY KEY	Unique police station ID.
`station_name`	String	NOT NULL	Official station name (e.g., "Sadar Police Station").
`phone_number`	String	NOT NULL	Official contact hotline number.
`address`	Text	NOT NULL	Physical street address in Nagpur.
`coordinates`	GeoJSON Point	SPATIAL INDEX	Exact latitude and longitude point location.
`is_24x7`	Boolean	DEFAULT TRUE	Flag indicating round-the-clock operation.

4. Hospitals & Emergency Medical Table (`hospitals`)

Field Name	Data Type	Constraints	Description
`hospital_id`	UUID	PRIMARY KEY	Unique hospital record ID.
`hospital_name`	String	NOT NULL	Name of facility (e.g., "GMC Nagpur Trauma Center").
`facility_type`	Enum	'HOSPITAL'|'PHARMACY'	Medical service type.
`emergency_contact`	String	NOT NULL	Direct emergency desk phone number.
`coordinates`	GeoJSON Point	SPATIAL INDEX	Lat/Lng position for proximity queries.
`has_trauma_unit`	Boolean	DEFAULT FALSE	Availability of dedicated emergency trauma care.

5. Verified News & Safety Updates Table (`news_updates`)

Field Name	Data Type	Constraints	Description
`news_id`	UUID	PRIMARY KEY	Unique news record identifier.
`headline`	String	NOT NULL	Article title or advisory headline.
`ai_summary`	Text	NULLABLE	Gemini AI-generated 2-bullet summary.
`source_name`	String	NOT NULL	Publisher name (e.g., "Nagpur Police Advisory").
`verification_status`	Enum	'VERIFIED'|'PENDING'	Strict verification flag.
`published_at`	Timestamp	NOT NULL	Publication timestamp.

6. User Feedback & Safety Reports (`user_feedback`)

Field Name	Data Type	Constraints	Description
`report_id`	UUID	PRIMARY KEY	Unique crowdsourced feedback ID.
`category`	Enum	'UNLIT'|'HAZARD'|'OTHER'	Category of community report.
`coordinates`	GeoJSON Point	NOT NULL	Reported location coordinates.
`description`	Text	NOT NULL	Anonymous issue description.
`status`	Enum	'SUBMITTED'|'VERIFIED'	Admin review status.

15 API Design & REST Endpoint Specifications

N-SHIELD exposes clean, RESTful JSON API endpoints for integration across web clients, mobile apps, and municipal data sinks:

Endpoint 1: GET /api/risk-zones

Purpose: Fetches GeoJSON polygon array of all defined safety risk zones in Nagpur.


```
{
  "success": true,
  "count": 3,
  "zones": [
    {
      "id": "zone-ngp-001",
      "zone_name": "Sitabuldi Market Zone",
      "risk_level": "ORANGE",
      "risk_score": 68,
      "coordinates": [[[79.082, 21.142], [79.091, 21.142], [79.091, 21.149], [79.082, 21.149]]]
    }
  ]
}
```

Endpoint 2: GET /api/location-safety

Purpose: Query safety insights, Gemini AI summaries, and nearby services for a latitude/longitude point.

Parameters: `lat=21.1458&lng=79.0882`


```
{
  "success": true,
  "location": "Sitabuldi Main Market",
  "risk_assessment": {
    "level": "ORANGE",
    "score": 68,
    "primary_factors": ["High Evening Density", "Moderate Patrols"]
  },
  "ai_summary": {
    "summary_bullets": [
      "High commercial activity during peak metro operating hours.",
      "Commuters recommended to stick to lit main station corridors late at night."
    ],
    "generated_at": "2026-09-09T14:30:00Z"
  }
}
```


Endpoint 3: GET /api/police-stations

Purpose: Fetch nearest police station details based on proximity.


```
{
  "success": true,
  "police_stations": [
    {
      "station_name": "Sitabuldi Police Station",
      "distance_km": 0.4,
      "phone": "0712-2522222",
      "address": "Near Station Road, Sitabuldi, Nagpur"
    }
  ]
}
```


Endpoint 4: GET /api/hospitals

Purpose: Fetch nearby trauma centers and emergency pharmacies.


```
{
  "success": true,
  "hospitals": [
    {
      "name": "Government Medical College & Hospital (GMC)",
      "facility_type": "HOSPITAL",
      "distance_km": 1.2,
      "emergency_phone": "0712-2744444",
      "trauma_available": true
    }
```

```
  ]
}
```

Endpoint 5: POST /api/feedback

Purpose: Submit anonymous crowdsourced community safety reports.

```
// Request Payload:
{
  "category": "UNLIT_STREET",
  "latitude": 21.1350,
  "longitude": 79.0620,
  "description": "Streetlights non-functional on 200m stretch near flyover pillar 14."
}

// Response:
{
  "success": true,
  "message": "Report logged successfully. Thank you for contributing to Nagpur safety.",
  "report_id": "FB-90234"
}
```

16 Innovation & Uniqueness

N-SHIELD stands apart from standard navigation apps and static government portals through six structural innovations:

Innovation Pillar	Standard Existing Tools	N-SHIELD Differentiation
1. Context Beyond Traffic	Optimize purely for travel distance and traffic delays.	Evaluates spatial lighting, emergency density, and safety incidents along corridors.
2. Plain-Language AI Summarization	Publish raw PDF tables of annual crime statistics.	Generates instant, easy-to-understand 2-bullet AI situation summaries for citizens.
3. Citizen-First Dashboard	Official portals built for administrative personnel.	Clean, modern mobile PWA designed for everyday commuters, students, and families.
4. Unified Emergency Directory	Contacts hidden across multiple non-responsive websites.	One-tap direct SOS calling for police, medical, fire, and helpline hotlines.
5. Responsible AI Governance	Aggressive algorithmic profiling or sensationalized news headlines.	Strict non-predictive, privacy-safe, transparent, anti-bias AI framework.
6. Localized Smart City Scalability	Generic global tools lacking hyper-local city focus.	Tailored specifically for Nagpur initially, built on a modular template ready for pan-India expansion.

17 Fund My Crazy Category Explanation

> **SELECTED CATEGORY: COMMUNITY & LIVING**
>
> N-SHIELD directly addresses the fundamental human need for safety, security, and peace of mind within urban living environments.

Why Community & Living?

Urban safety is not an abstract statistical concept—it is a daily lived reality. Every single day, thousands of students, late-shift IT workers, healthcare personnel, and families in Nagpur make decisions about when to travel, which road to take, and where to walk. By providing transparent, accessible safety intelligence, N-SHIELD directly elevates the quality of community living, fosters public trust, and promotes urban mobility.

Why Now? The Convergence of Tech & Urban Growth

- **Rapid Smart City Growth:** Nagpur is undergoing rapid infrastructure expansion (Metro Rail, MIHAN IT Hub, Ring Road developments). Citizens urgently require smart safety tools matching this growth.
- **Ubiquitous Smartphone & 5G Access:** High mobile data penetration in tier-2 Indian cities enables instant delivery of location-based mapping tools to citizens on the go.
- **Breakthroughs in Generative AI:** AI models like Google Gemini can now summarize scattered public news feeds accurately in real-time, making automated synthesis viable for the first time.
- **Demand for Transparent Civic Data:** Modern Indian citizens expect smart city initiatives to deliver practical, direct-to-citizen digital services.

18 Future Scope & Roadmap

Planned Technological Innovations

1. **Safer Route Navigation Engine:** Full integration of risk polygons with Google Directions API to automatically calculate and highlight the safest alternative travel path.
2. **Real-Time Police & Municipal Portal Integration:** Official API data hooks with Nagpur Municipal Corporation (NMC) and Police Control Rooms for live verified advisories.
3. **Multi-Language Accessibility Support:** Native localization in **English, Hindi (हिंदी), and Marathi (मराठी)** to serve all demographic segments in Nagpur.
4. **Moderated Crowdsourced Reporting:** Citizen reporting portal for unlit streetlights, road hazards, or low-patrol zones with automated spam filtering and admin moderation.
5. **Native Mobile Applications (Android & iOS):** Packaging PWA codebase into Flutter/React Native builds with background push notification support for urgent area alerts.
6. **Smartwatch Emergency SOS Companion App:** One-touch SOS trigger for Wear OS and Apple Watch devices transmitting real-time GPS coordinates to emergency contacts.
7. **Expanded City Coverage:** Cloning N-SHIELD framework to deploy for Pune, Nashik, Chhatrapati Sambhajinagar, Thane, and Mumbai metro regions.

8. **Advanced Time-Series Incident Analysis:** Long-term trend analysis models identifying seasonal or hourly safety patterns to assist civic planning.
9. **Off-Grid SMS Emergency Trigger:** SMS fallback integration allowing citizens to fetch nearest police station contacts even without active internet connectivity.

19 Limitations & Technical Constraints

In the interest of academic integrity and engineering rigor, the current version of N-SHIELD operates within the following defined constraints:

Limitation Area	Technical & Operational Reality
1. Public Data Dependency	Accuracy of risk zones depends directly on the availability, timeliness, and granularity of public incident reports and news feeds.
2. Non-Predictive Boundary	System provides spatial risk guidance based on historical and aggregated data; it cannot predict isolated, random individual crimes.
3. Official Response Reliance	N-SHIELD facilitates rapid dialing to emergency services but does not control official emergency dispatch response times.
4. News Verification Overhead	To prevent misinformation, news updates require automated or admin verification before map publishing, introducing a minor ingestion lag.
5. GPS Connectivity Requirement	Interactive mapping and location-based proximity queries require active client GPS and internet connection.

20 Conclusion

N-SHIELD (AI-Powered Urban Safety Intelligence) represents a transformative approach to smart city safety management for Nagpur. By bridging the gap between scattered public safety data, spatial mapping technologies, and responsible generative AI summarization, N-SHIELD places actionable safety intelligence directly into the hands of everyday citizens.

Whether assisting a student commuting late from university, a woman professional navigating evening transit from MIHAN IT Park, or a visitor arriving at Nagpur Railway Station, N-SHIELD delivers rapid clarity, essential service access, and peace of mind. Built upon modern web standards (Next.js, TypeScript, Google Maps API, Gemini AI) and guided by strict privacy and anti-bias principles, N-SHIELD serves as a powerful, scalable blueprint for urban safety across Nagpur, Maharashtra, and India.

N-SHIELD: Empowering Nagpur Citizens Through Intelligent Safety Science

A Glossary of Technical & Domain Terms

- **GeoJSON:** An open standard format for encoding a variety of geographic data structures using JavaScript Object Notation.
- **Spatial Indexing:** Database optimization techniques (e.g., PostGIS R-Tree) that allow high-speed spatial queries like "find points within polygon".
- **Gemini 1.5 Flash:** Lightweight, low-latency multimodal LLM developed by Google optimized for fast text summarization.
- **PWA (Progressive Web App):** Web applications built using modern web APIs to deliver app-like experiences on desktop and mobile.
- **SSR (Server-Side Rendering):** Technique where HTML pages are generated on the server for each request, delivering faster initial page loads.
- **Click-to-Call (tel:):** URI scheme enabling mobile browsers to initiate cellular phone calls directly from webpage buttons.
- **Risk Zone Polygon:** A series of connected GPS coordinate points forming a bounded geographic area on a map representing a specific risk level.

B References & Developer Resources

1. **Google Maps JavaScript API Documentation:** `https://developers.google.com/maps/documentation/javascript`
2. **Next.js Framework & App Router Architecture:** `https://nextjs.org/docs`
3. **Google Gemini API Documentation:** `https://ai.google.dev/docs`
4. **MongoDB Spatial & Geospatial Queries:** `https://www.mongodb.com/docs/manual/geospatial-queries/`
5. **PostgreSQL + PostGIS Spatial Database Extension:** `https://postgis.net/documentation/`
6. **Nagpur Municipal Corporation (NMC) Official Portal:** `https://www.nmcnagpur.gov.in/`
7. **Nagpur City Police Portal & Advisories:** `https://nagpurpolice.gov.in/`
8. **Tailwind CSS Documentation:** `https://tailwindcss.com/docs`

N-SHIELD

AI-Powered Urban Safety Intelligence

NAGPUR, MAHARASHTRA

Prepared by: Khushal Deogade

THE PROBLEM

Fragmented safety data across Nagpur makes it difficult for citizens to understand neighborhood safety before traveling. Navigation apps ignore safety factors like lighting and emergency access.

THE SOLUTION

An interactive smart city safety dashboard combining Google Maps spatial risk layers (Green/Orange/Red), one-tap emergency services finder, verified news feeds, and Gemini AI safety summaries.

KEY HIGHLIGHTS & ARCHITECTURE

Tech Stack	Next.js, TypeScript, Tailwind CSS, Google Maps API, Gemini AI, MongoDB/Postgres, Vercel
Target Users	Students, Women Commuters, Night Shift Professionals, Senior Citizens, Tourists, Families
Core Features	Risk Map, Category Filters, Emergency Locator (Police/Hospitals), One-Tap SOS, AI Summaries
Responsible AI	Strictly non-predictive, privacy-safe (zero PII storage), anti-bias, transparent source tracking

FUND MY CRAZY COMPETITION IMPACT STATEMENT

Selected Category: **Community & Living**. N-SHIELD converts complex, scattered public safety signals into intuitive daily spatial intelligence, fostering civic trust, safe mobility, and community resilience in Nagpur with immediate scalability across India.